9 Prime and Maximal Ideals

In this section all rings are **commutative** with an identity element $1 \neq 0$.

The sign $\subsetneq$ means contained and not equal.

Let R be a ring. An ideal I of R is called **proper** if $I \subsetneq R$.

Definition 9.1 (Maximal Ideal)**.**
A proper ideal $M \subsetneq R$ is called a *maximal* ideal if the only ideals of R containing M are M and R, i.e. if $M \subset I$ and I is an ideal then $M = I$ or $I = R$.

Maximal ideals are maximal elements within the set of proper ideals for the relation of inclusion.

Proposition 9.1.
Let $J \subsetneq R$ be a proper ideal. There exists a maximal ideal M of R containing J: $J \subset M$.

Proof. Let $\mathcal{S}$ be the set of all proper ideals of R containing J. The set $\mathcal{S}$ is nonempty (as it contains J) and (partially) ordered by the relation of inclusion. Let $J \subset \ldots I_k \subset I_{k'} \subset \ldots$, with $k, k', \ldots \in K$: an index set, be a chain of proper ideals containing J. By Zorn's Lemma we need to show that $\{I_k\}_{k\in K}$ has an upper bound. Let $\mathcal{I} = \bigcup_{k\in K} I_k$. Then $\mathcal{I}$ is an ideal of R: if $a, b \in \mathcal{I}$ then $a \in I_k, b \in I_{k'}$ for some $k, k' \in K$. Since the family $\{I_k\}_{k\in K}$ is a chain either $I_k \subset I_{k'}$ or $I_{k'} \subset I_k$, say $I_k \subset I_{k'}$, hence $a + b, ab \in I_{k'} \in \mathcal{I}$. Similarly if $a \in \mathcal{I}$, $r \in R$, $\exists k \in K$ with $a \in I_k$, hence $ra \in I_k \subset \mathcal{I}$ since I_k is an ideal. Further $1 \notin \mathcal{I}$, for otherwise $\exists k \in K$ with $1 \in I_k$ hence $I_k = R$ which is not the case since I_k is a proper ideal. Clearly $\mathcal{I}$ is an upper bound for $\{I_k\}_{k\in K}$. □

In particular, applying Proposition 9.1 to the zero ideal $\{0_R\}$ of R we find that maximal ideals always exist.

Theorem 9.2 (Lattice Isomorphism Theorem)**.**
Let $J \subset R$ be an ideal. There is a one to one correspondence between the set of ideals I of R containing J and the set of ideals of the factor ring R/J.

Proof. Let $\mathcal{S}$ be the set of ideals I of R such that $J \subset I$. Let $\mathcal{T}$ be the set of ideals of R/J. It suffices to construct a bijection from $\mathcal{S}$ to $\mathcal{T}$.

For $I \in \mathcal{S}$, define I/J to be the set $\{\bar{b} \in R/J : \bar{b} = \bar{a}, \text{for some } a \in I\}$. If $\bar{b} \in I/J$, then $\bar{b} = \bar{a}$, for some $a \in I$. So $b - a \in J \subset I$. As $a \in I$, $b \in I$. Thus, $\bar{b} \in I/J$ if and only if $b \in I$.

Moreover, $I/J \in \mathcal{T}$. Indeed, $0 \in I$, so $\bar{0} \in I/J$. Suppose $\bar{a}, \bar{b} \in I/J$, so $a \in I$ and $b \in I$. Then $\bar{a} + \bar{b} = \overline{a + b}$ and $a + b \in I$, so $\bar{a} + \bar{b} \in I/J$. Likewise, $-a \in I$ so $-\bar{a} = \overline{-a} \in I/J$, as $-a \in I$. If $\bar{r} \in R/J$, then $ar \in I$, whence $\bar{a}\bar{r} \in I/J$.

Let $\phi : R \to R/J$ be the homomorphism $\phi(a) = \bar{a}$. The preceding comments show that $\phi(I) = I/J \in \mathcal{T}$ for each $I \in \mathcal{S}$. Define $f : \mathcal{S} \to \mathcal{T}$ by $f(I) = \phi(I)$. It remains to construct a map $g : \mathcal{T} \to \mathcal{S}$ that is an inverse of f.

Let $\widetilde{I} \in \mathcal{T}$ be an ideal of R/J. Then, $\phi^{-1}(\widetilde{I})$ is an ideal of R. Further, $J \subset \phi^{-1}(\widetilde{I})$ since if $a \in J$ then $\phi(a) = \bar{a} = \bar{0} \in \widetilde{I}$. Thus there is map $g : \mathcal{T} \to \mathcal{S}$ given by $g(\widetilde{I}) = \phi^{-1}(\widetilde{I})$.

It is left as an exercise to show that $g(f(I)) = I$ for all $I \in \mathcal{S}$ and that $f(g(\widetilde{I})) = \widetilde{I}$, for all $\widetilde{I} \in \mathcal{T}$. From this, it follows that f is a one-to-one correspondence between the ideals of R that contain J and the ideals of R/J. □

Theorem 9.3.
The ideal $M \subsetneq R$ is a maximal ideal if and only if the quotient ring R/M is a field.

Proof. By the Lattice Isomorphism Theorem there is a one to one correspondence between the set of ideals I of R containing M and the ideals of the quotient ring R/M. Now R/M is a field if and only if the only ideals of R/M are R/M and $\{\bar{0}\}$ [this is essentially the content of CW4], in which case the corresponding ideals of R containing M are: $\phi^{-1}(R/M) = R$ and $\phi^{-1}(0_{R/M}) = \mathrm{Ker}(\phi) = M$; where $\phi : R \to R/M$ is the map $a \mapsto \phi(a) = \bar{a}$, i.e. M is a maximal ideal. □

Example 9.1.

1. Let p be a prime number. The ideal $p\mathbb{Z} = (p)_{\mathbb{Z}} = \{pk : k \in \mathbb{Z}\}$ is a maximal ideal of $\mathbb{Z}$ since the factor ring $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$ is a field (see Examples 7.5).
2. The ideal $I = (X)_R = \{Xf(X) : f \in \mathbb{Z}[X]\}$ is **NOT** a maximal ideal of $R = \mathbb{Z}[X]$ since $(X)_R \subsetneq (2, X)_R \subsetneq \mathbb{Z}[X]$.
3. Let $R = \{f : [0, 1] \to \mathbb{R}\}$ be the set of all functions from $[0, 1]$ to $\mathbb{R}$. then R is a ring for the operations $+$, $\times$, defined by $(f + g)(x) = f(x) + g(x)$ and $(f \times g)(x) = f(x)g(x)$. Let $a \in \mathbb{R}$ and $M_a = \{f \in R : f(a) = 0\}$, then M_a is an ideal of R (check this fact). The map $\phi : R \to \mathbb{R}$ defined by $f \mapsto \phi(f) = f(a)$ is a surjective ring homomorphism (check this fact) and $\mathrm{Ker}(\phi) = M_a$. The FIT implies R/M_a is isomorphic to $\mathbb{R}$ which is a field hence R/M_a is a field and M_a is a maximal ideal.
4. If $\phi : R \to F$ is a surjective homomorphism from a ring R to a field F with kernel I then I is a maximal ideal. Indeed by the FIT: R/I is isomorphic to the field F hence R/I itself is a field (check this fact) and I is a maximal ideal.
5. The ideal
$$M = (2, X) = \{2f(X) + Xg(X) : f, g \in \mathbb{Z}[X]\}$$
is a maximal ideal of $\mathbb{Z}[X]$. Indeed the map
$$\phi : \mathbb{Z}[X] \to \mathbb{F}_2 = \{\bar{0}, \bar{1}\}$$
defined by
$$\phi(a_0 + a_1X + \ldots + a_nX^n) = \bar{a}_0$$
is a surjective homomorphism (check this fact) with kernel M. Indeed $\phi(2) = \phi(X) = \bar{0}$ hence $\phi(2f(X) + Xg(X)) = \phi(2)\phi(f(X)) + \phi(X)\phi(g(X)) = 0$ and $M \subset \mathrm{Ker}\,\phi$.
If $f(X) = a_0 + a_1X + \ldots + a_nX^n \in \mathbb{Z}[X]$ with $\phi(f(X)) = \bar{a}_0 = \bar{0}$ [i.e. $f(X) \in \mathrm{Ker}(\phi)$] then $a_0 = 2b$ and $f(X) = 2b + X(a_1 + a_2X + \ldots + a_nX^{n-1}) \in M$.
The FIT implies $\mathbb{Z}[X]/M$ is isomorphic to $\mathbb{F}_2$: a field, hence $\mathbb{Z}[X]/M$ is a field and M is a maximal ideal.
6. Let $R = \mathbb{Z}[i]$, we show the ideal
$$I = (1 + i)_R = \{(1 + i)(a + bi) : a, b \in \mathbb{Z}\}$$
is a maximal ideal of R by showing the existence of a surjective homomorphism from R to a field with kernel I. Let
$$\phi : \mathbb{Z}[i] \to \mathbb{F}_2 = \{\bar{0}, \bar{1}\}$$
be the map defined by
$$\phi(a + ib) = \overline{a + b}$$

The map ϕ is surjective as $\bar{0} = \phi(1+i)$ and $\bar{1} = \phi(1+2i)$. Further ϕ is a ring homomorphism:

$$\begin{aligned}\phi[(a+ib)+(c+id)] &= \phi[(a+c)+i(b+d)] = \overline{(a+c)+(b+d)} \\ &= \overline{(a+b)+(c+d)} = \overline{(a+b)} + \overline{(c+d)} \\ &= \phi(a+bi)+\phi(c+di).\end{aligned}$$

and

$$\begin{aligned}\phi[(a+ib)(c+id)] &= \phi[(ac-bd)+i(ad+bc)] = \overline{ac-bd+ad+bc} \\ &= \overline{ac+bd+ad+bc} = \overline{(a+b)(c+d)} \\ &= \overline{(a+b)}\,\overline{(c+d)} = \phi(a+ib)\phi(c+id).\end{aligned}$$

[for the third equality we used the fact that $\overline{-1} = \bar{1}$ in $\mathbb{F}_2$]. We show $\mathrm{Ker}(\phi) = (1+i)_R$. Clearly $(1+i)_R \subset \mathrm{Ker}(\phi)$ since $\phi(1+i) = \overline{1+1} = \bar{0}$, hence $\phi((1+i)(a+ib)) = \phi(1+i)\phi(a+ib) = \bar{0}\phi(a+ib) = \bar{0}$ for every element $(1+i)(a+ib) \in I$.
Reciprocally let $a+ib \in \mathrm{Ker}(\phi)$. Then $\phi(a+ib) = \overline{a+b} = \overline{a-b} = \bar{0}$ hence $a = 2k+b$ for some integer k and $a+ib = 2k+b+ib = (1+i)(1-i)k + b(1+i) = (1+i)(b+k-ki) \in I$ as required, here we used $2 = (1+i)(1-i)$ in $\mathbb{Z}[i]$.
The FIT implies $\mathbb{Z}[i]/\mathrm{Ker}(\phi) = \mathbb{Z}[i]/I$ is isomorphic to $\mathbb{F}_2$: a field, hence $\mathbb{Z}[X]/I$ is a field and I is a maximal ideal.

Definition 9.2 (Prime Ideal).
A proper ideal $P \subsetneq R$ is called a *prime* ideal if whenever the product ab of two elements $a, b \in R$ is an element of P then either a or b is an element of P:

$$\{ab \in P\} \Longrightarrow \{a \in P \text{ or } b \in P\}.$$

Example 9.2.
If p is a prime number the ideal $p\mathbb{Z} = (p)_{\mathbb{Z}} = \{pk : k \in \mathbb{Z}\}$ is a prime ideal. indeed let $a, b \in \mathbb{Z}$ with $ab \in p\mathbb{Z}$. Thus $ab = pk$ for some $k \in \mathbb{Z}$ and p divides ab. Since p is prime either p divides a which implies $a \in p\mathbb{Z}$ or p divides b which implies $b \in p\mathbb{Z}$.

Theorem 9.4.
The ideal $P \subsetneq R$ is a prime ideal if and only if the factor ring R/P is an integral domain.

Proof. Let $a, b \in R$ with $ab \in P$ and consider the homomorphism

$$\phi : R \to R/P$$

defined by

$$r \mapsto \phi(r) = \bar{r}.$$

Then $\overline{ab} = \bar{a}\bar{b} = \bar{0}_R$ since $ab \in P$. Now P is a prime ideal if and only if whenever $ab \in P$, which is equivalent $\overline{ab} = \bar{a}\bar{b} = \bar{0}_R$, then either $a \in P$ which is equivalent $\bar{a} = \bar{0}_R$, or $b \in P$ which is equivalent $\bar{b} = \bar{0}_R$, this latter condition is equivalent to R/P being an integral domain. □

Example 9.3.

1. In an integral domain R the ideal $\{0_R\}$ is a prime ideal since whenever $ab = 0_R$ then either $a = 0_R$ or $b = 0_R$.
2. In $\mathbb{Z}$ the ideals $\{0\}$ and $\{p\mathbb{Z}\}_p$ for p a prime number are prime ideals. The ideal $\{0\}$ is not a maximal ideal since $\{0\} \subsetneq m\mathbb{Z} \subsetneq \mathbb{Z}$ for any nonzero integer m which is different from both 1 and -1.
3. If $\phi : R \to D$ is a surjective homomorphism from a ring R to an integral domain D with kernel J then J is a prime ideal. Indeed by the FIT: R/J is isomorphic to the integral domain D hence R/J itself is an integral domain (prove this) and J is a prime ideal.
4. The ideal $(X)_{\mathbb{Z}[X]} = \{Xf(X) : f(X) \in \mathbb{Z}[X]\}$ of $\mathbb{Z}[X]$ is a prime ideal. Indeed consider the map
$$\phi : \mathbb{Z}[X] \to \mathbb{Z}$$
defined by
$$\phi(a_0 + a_1X + \ldots + a_nX^n) = a_0.$$
Then ϕ is a surjective homomorphism (check this fact) with kernel
$$\mathrm{Ker}(\phi) = \{f(X) = a_0 + a_1X + \ldots + a_nX^n : a_0 = 0\} = (X)_{\mathbb{Z}[X]}$$
and $(X)_{\mathbb{Z}[X]}$ is a prime ideal since $\mathbb{Z}$ is an integral domain. Note that the ideal $(X)_{\mathbb{Z}[X]}$ is prime but not maximal (see Examples 9.1).

A prime ideal need not be a maximal ideal [see Examples 9.3 and (4)]. However the converse holds.

Proposition 9.5.
A maximal ideal is a prime ideal.

Proof. Indeed if M is a maximal ideal of R then R/M is a field hence is an integral domain (since every field is an integral domain, see §7), and M is a prime ideal by Theorem 9.4. □

9.1 Field of Fractions

Let R be an integral domain and
$$\widetilde{F} = \{(a, b) : a, b \in R, b \neq 0_R\}.$$
Define the relation $\mathcal{H}$ among elements of F by
$$(a, b)\ \mathcal{H}\ (c, d) \iff ad = bc.$$
The relation $\mathcal{H}$ is an equivalence relation (check this fact). Let $F = \widetilde{F}/\mathcal{H}$ be the set of equivalence relations of $\mathcal{H}$. We will denote the equivalence class (a, b) by the fraction $\frac{a}{b}$. Thus
$$F = \left\{\frac{a}{b} :\ a, b \in R,\ b \neq 0_R\right\},$$
and
$$\frac{a}{b} = \frac{c}{d} \iff ad = bc.$$
Define the operations $+$ and $\times$ on the set F by
$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \qquad \text{and} \qquad \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}.$$

Note that $bd \neq 0_R$ since $b \neq 0_R$, $d \neq 0_R$, and R is integral. Hence both $\frac{ad+bc}{bd}$ and $\frac{ac}{bd}$ are elements of F.

These operations are well defined and do not depend on the choice of a representative of the class of equivalence of (a, b).

Indeed assume $(a, b)\ \mathcal{H}\ (a', b') \iff ab' = ba'$, and $(c, d)\ \mathcal{H}\ (c', d') \iff cd' = dc'$, then

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} = \frac{a'd' + b'c'}{b'd'} = \frac{a'}{b'} + \frac{c'}{d'}$$

since $(ad + bc)b'd' = adb'd' + bcb'd' = ba'db' + bb'c'd = bd(a'd' + b'c')$, and similarly (check)

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} = \frac{a'c'}{b'd'} = \frac{a'}{b'} \times \frac{c'}{d'}.$$

The set F endowed with the operations $+$ and $\times$ is a commutative ring (check this fact) with identity

$$1_F = \frac{1_R}{1_R} \qquad 0_F = \frac{0_R}{1_R} \qquad \text{and} \qquad -(\frac{a}{b}) = \frac{-a}{b}.$$

Further F is a field: if $\frac{a}{b} \neq 0_F$ then $a \neq 0_R$ $[\frac{a}{b} = 0_F = \frac{0_R}{1_R} \iff a = a1_R = b0_R = 0_R]$

$$\frac{a}{b} \times \frac{b}{a} = \frac{ab}{ab} = \frac{1_R}{1_R} = 1_F,$$

and $\frac{b}{a}$ is the inverse of $\frac{a}{b}$. The field F is called the **field of fraction** of R and denoted $\mathrm{Fr}(R)$.

Example 9.4.

1. $\mathbb{Z}$ is an integral domain and $\mathrm{Fr}(\mathbb{Z}) = \mathbb{Q}$.
2. Let K be a field. The ring $K[X]$ is an integral domain, its field of fraction is denoted $K(X) = \mathrm{Fr}(K[X])$ and called the ring of rational functions in the variable X. Thus

$$K(X) = \Big\{\frac{f(X)}{g(X)} : f, g \in K[X], g \neq 0\Big\}$$

with

$$\frac{f(X)}{g(X)} + \frac{h(X)}{q(X)} = \frac{f(X)q(X) + g(X)h(X)}{g(X)q(X)} \qquad \text{and} \qquad \frac{f(X)}{g(X)} \times \frac{h(X)}{q(X)} = \frac{f(X)h(X)}{g(X)q(X)}.$$

If $\frac{f(X)}{g(X)} \in K(X)$ is nonzero then $f(X) \neq 0$ and the inverse of $\frac{f(X)}{g(X)}$ is $\frac{g(X)}{f(X)}$:

$$\frac{f(X)}{g(X)} \times \frac{g(X)}{f(X)} = \frac{f(X)g(X)}{f(X)g(X)} = \frac{1_K}{1_K} = 1_{K(X)}.$$

Proposition 9.6.
Let R be an integral domain. Then the followings hold.

(i) *The map $\psi : R \to \mathrm{Fr}(R)$ defined by $r \mapsto \psi(r) = \frac{r}{1_R}$ is an injective ring homomorphism.* ***(ii)*** *Let $\phi : R \to K$ be an injective homomorphism into a field K. There exists a unique injective homomorphism $\tau : \mathrm{Fr}(R) \to K$ with the property*

$$\tau(\psi(r)) = \tau(\frac{r}{1_R}) = \phi(r)$$

for every $r \in R$.

Proof. **(i)**

$$\psi(r+s) = \frac{r+s}{1_R} = \frac{r}{1_R} + \frac{s}{1_R} = \psi(r) + \psi(s)$$

and

$$\psi(rs) = \frac{rs}{1_R} = \frac{r}{1_R} \times \frac{s}{1_R} = \psi(r)\psi(s).$$

Further if $\psi(r) = \frac{r}{1_R} = \frac{0_R}{1_R}$ then $r = r \times 1_R = 0_R \times 1_R = 0_R$ and ψ is injective.

(ii) Define $\tau : \mathrm{Fr}(R) \to K$ by

$$\tau(\frac{r}{s}) = \phi(r)\phi(s)^{-1}$$

[which makes sense since $s \neq 0_R$, $\phi(s) \neq 0_K$ as ϕ is injective, and the inverse $\phi(s)^{-1}$ of $\phi(s)$ exists in K]. Then

$\tau(\frac{r}{s} + \frac{t}{h}) = \tau(\frac{rh+st}{sh}) = \phi(rh+st)\phi(sh)^{-1} = [\phi(r)\phi(h) + \phi(s)\phi(t)]\phi(s)^{-1}\phi(h)^{-1} = \phi(r)\phi(s)^{-1} + \phi(t)\phi(h)^{-1} = \tau(\frac{r}{s}) + \tau(\frac{t}{h})$,

further

$\tau(\frac{r}{s}\frac{t}{h}) = \tau(\frac{rt}{sh}) = \phi(rt)\phi(sh)^{-1} = [\phi(r)\phi(t)]\phi(s)^{-1}\phi(h)^{-1} = \phi(r)\phi(s)^{-1}\phi(t)\phi(h)^{-1} = \tau(\frac{r}{s})\tau(\frac{t}{h})$, hence τ is a homomorphism.

and

$\tau(1_{\mathrm{Fr}(R)}) = \tau(\frac{1_R}{1_R}) = \phi(1_R)\phi(1_R)^{-1} = 1_K(1_K)^{-1} = 1_K$.

If $\tau(\frac{r}{s}) = \phi(r)\phi(s)^{-1} = 0_K$ then $\phi(r) = 0_K$ [since K is integral and $\phi(s)^{-1} \neq 0_K$] hence $r = 0_R$ since ϕ is injective. This shows τ is injective.

The property $\tau(\psi(r)) = \tau(\frac{r}{1_R}) = \phi(r)$ determines τ uniquely since $\tau(\frac{r}{s}) = \tau(\psi(r)\psi(s)^{-1}) = \tau(\psi(r))\tau(\psi(s)^{-1}) = \tau(\psi(r))\tau(\psi(s))^{-1} = \phi(r)\phi(s)^{-1}$. □

9.2 The Chinese Remainder Theorem

Given two rings R_1 and R_2 we define their **product**

$$R_1 \times R_2 = \{(r_1, r_2) : r_1 \in R_1, r_2 \in R_2\}$$

and endow it with the operations + and × defined by

$$(r_1, r_2) + (s_1, s_2) = (r_1 + s_1, r_2 + s_2) \qquad (r_1, r_2) \times (s_1, s_2) = (r_1 s_1, r_2 s_2).$$

The set $R_1 \times R_2$ endowed with + and × is a commutative ring (check this fact) with identity and zero elements:

$$1_{R_1 \times R_2} = (1_{R_1}, 1_{R_2}) \qquad \text{and} \qquad 0 = (0_{R_1}, 0_{R_2}) \qquad \text{respectively.}$$

Similarly as above we can define the product of any given set of rings with addition and multiplication defined component-wise.

- Two ideals of a ring R are called **co-maximal** if $I + J = R$. If I, J are co-maximal ideals of R then $IJ = I \bigcap J$ [see Exercise 7(d) in Problem Sheet 4]

Theorem 9.7 (Chinese Remainder Theorem)**.**
Let I, J be ideals of R. The map

$$\phi : R \to R/I \times R/J$$

defined by

$$r \mapsto (r + I, r + J)$$

is a ring homomorphism with kernel $I \bigcap J$.

If I and J are co-maximal then this map is surjective and $I \bigcap J = IJ$, in this case $R/IJ = R/(I \bigcap J)$ is isomorphic to $R/I \times R/J$.

Proof. We have

$$\phi(r+s) = (r+s+I, r+s+J) = (r+I+s+I, r+J+s+J) = (r+I, r+J)+(s+I, s+J) = \phi(r)+\phi(s)$$

$$\phi(rs) = (rs + I, rs + J) = ((r + I)(s + I), (r + J)(s + J)) = (r + I, r + J)(s + I, s + J) = \phi(r)\phi(s)$$

and

$$\phi(1_R) = (1_R + I, 1_R + J) = 1_{(R/I\times R/J)}$$

hence ϕ is a homomorphism.

$\mathrm{Ker}(\phi) = \{r \in R : (r + I, r + J) = (0_{R/I}, 0_{R/J})\} = \{r \in R : r \in I, r \in J\} = I \bigcap J.$

Suppose $I + J = R$, we show ϕ is surjective in this case.

Let $(r_1 + I, r_2 + J) \in R/I \times R/J$. There exists $a \in I, b \in J$ with

$$1 = a + b$$

(since $I + J = R$). In particular $a = 1 - b$ and

$$a + J = (1 - b) + J = 1 + J - (b + J) = 1 + J,$$

$b = 1 - a$ and

$$b + I = (1 - a) + I = 1 + I - (a + I) = 1 + I.$$

Consider the element

$$r = r_2a + r_1b \in R$$

then

$$\phi(r) = ((r_2a + r_1b) + I, (r_2a + r_1b) + J) = (r_2a + I + r_1b + I, r_2a + J + r_1b + J) = (r_1b + I, r_2a + J)$$

$$= (r_1 + I, r_2 + J)(b + I, a + J) = (r_1 + I, r_2 + J)(1 + I, 1 + J) = (r_1 + I, r_2 + J).$$

Further $\mathrm{Ker}(\phi) = I \bigcap J = IJ$ [see Exercise 8(d) in Problem Sheet 4] and the FIT for rings implies that $R/\mathrm{Ker}(\phi) = R/(I \bigcap J) = R/IJ$ is isomorphic to $R/I \times R/J$. □

Example 9.5.

- Let $n = p_1^{a_1} p_2^{a_2}$ be an integer with $p_1 \neq p_2$ primes. Then $\mathbb{Z} = p_1^{a_1}\mathbb{Z} + p_2^{a_2}\mathbb{Z}$, and the ideals $(p_1^{a_1})$ and $(p_1^{a_1})$ of $\mathbb{Z}$ are co-maximal. The Chinese Remainder Theorem implies that $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to the the product $\mathbb{Z}/p_1^{a_1}\mathbb{Z} \times \mathbb{Z}/p_2^{a_2}\mathbb{Z}$

- If $I_1, I_1, \ldots, I_n$ are ideals of R which are pairwise co-maximal: $I_i + I_j = R$ for $1 \leq i < j \leq n$, then the Chinese Remainder Theorem generalises as follows.
  The map
  $$\phi : R \to R/I_1 \times R/I_2 \times \ldots \times R/I_n$$
  defined by
  $$r \mapsto (r + I_1, r + I_2, \ldots, , r + I_n)$$
  is a surjective ring homomorphism with kernel
  $$I_1 \bigcap I_2 \bigcap \ldots \bigcap I_n = I_1 I_2 \ldots I_n$$
  and $R/(I_1 I_2 \ldots I_n) = R/(I_1 \bigcap \ldots \bigcap I_n)$ is isomorphic to $R/I_1 \times R/I_2 \times \ldots \times R/I_n$. Here the ideal $I_1 I_2 \ldots I_n$ consists of elements of R that are finite sums
  $$\sum a_1 a_2 \ldots a_n$$
  where $a_i \in I_i$ for $1 \leq i \leq n$.
  In particular if $m = p_1^{a_1} p_2^{a_2} \ldots p_n^{a_n}$ is an integer with $p_1, p_2, \ldots, p_n$ distinct primes, then the ideals $(p_i^{a_i})$ and $(p_j^{a_j})$ of $\mathbb{Z}$ are co-maximal for $1 \leq i < j \leq n$, and the Chinese Remainder Theorem states that $\mathbb{Z}/m\mathbb{Z}$ is isomorphic to the product $\mathbb{Z}/p_1^{a_1}\mathbb{Z} \times \mathbb{Z}/p_2^{a_2}\mathbb{Z} \times \ldots \times \mathbb{Z}/p_n^{a_n}\mathbb{Z}$.